Group 8

FREQUENT ITEMSET MINING ON SHOPPING BEHAVIOR USING ECLAT

LEE YIN SHEN A23CS0236 ELIJAH SHE YU SHENG A23CS0073 NEO LI XIN A23CS0253

(OVERVIEW)
ECLAT ALGORITHM

- The Eclat algorithm (Equivalence Class Transformation) is an efficient method for finding frequent itemsets in association rule mining. It works using the vertical data format, where the dataset is represented by a list of item identifiers for each customer purchases.
- Instead of scanning the dataset for frequent itemsets as in Apriori, Eclat uses set intersections to find frequ[ent itemsets, which improves performance for large datasets.

(ECLAT)
VERTICAL DATA FORMAT

- Vertical data format is used to represent a dataset in a way that each item is associated with a list of customers. Instead of the horizontal format (where customers are rows), each item is a column, and the customers are stored as sets of customer IDs (CIDs) that contain that item. This structure allows faster intersection of itemsets, improving algorithm efficiency.

(ECLAT)
EXAMPLE

Dataset:
- C1: {Pants, Coat, Shirt}
- C2: {Dress, Coat, Pants}
- C3: {Dress, Coat, Pants}

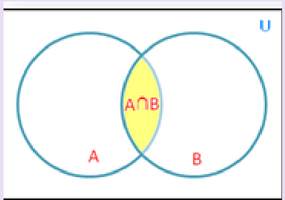
Vertical format representation:
- Shirt: {C1}
- Pants: {C1, C2, C3}
- Coat: {C1, C2, C3}
- Dress: {C2, C3}

Items	Customer ID
Shirt	C1
Pants	C1, C2, C3
Coat	C1, C2, C3
Dress	C2, C3

Eclat intersects the customer purchased itemset lists to find frequent itemsets, such as {Pants, Coat, Shirt} and {Dress, Coat, Pants}.

(ECLAT)
MINING FREQUENT ITEMSETS

- Eclat mines frequent itemsets by performing intersections between customer purchase lists. This allows it to efficiently identify itemsets that appear frequently in the dataset. The process involves finding subsets of customer purchase lists where multiple items co-occur, thus reducing the candidate set size compared to horizontal approaches.



A and B Customer Purchased Itemsets List

(ECLAT)
ADVANTAGES

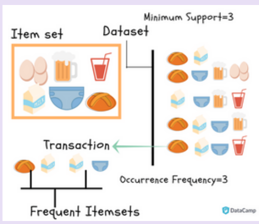
- **Non-intrusive:** Reduces complexity and cost by using aggregated data instead of requiring sensors to be installed on every device.
- **Efficient Load Disaggregation:** Energy monitoring and management can benefit from its ability to efficiently distinguish between the usage of particular appliances and overall load profiles.
- **Scalable:** Can be modified for use in energy monitoring systems at various scales, such as commercial, industrial, or domestic.
- **Supports Energy Optimization:** Assists users or systems in locating appliances that use a lot of energy and in optimising use to save costs and increase energy efficiency.

(ECLAT)
DISADVANTAGES

- **Accuracy Depends on Signature Uniqueness:** Accuracy may be decreased by misclassifying or confusing appliances with similar energy usage trends.
- **Requires Training Data:** For initial training, eclat might require labelled datasets, which aren't always available, like many machine learning-based techniques.
- **Limited Real-Time Performance:** Real-time analysis may be hampered by the computational complexity, particularly on devices with limited resources.
- **Performance Degrades with Noise:** Accurate categorisation may be impacted by its sensitivity to noise or variations in the power data.

(ECLAT)
APPLICATION OF ECLAT ALGORITHM MINING

- **Market Basket Analysis**: Identifying frequently purchased items together.
- **Recommendation Systems**: Suggesting products based on past purchase patterns.
- **Fraud Detection**: Discovering frequent patterns in fraudulent activities.



(ECLAT)
CHALLENGES IN ECLAT ALGORITHM MINING

- **Data Sparsity:** The algorithm's performance can degrade if the dataset has many rare items or sparse itemsets.
- **High Dimensionality:** Eclat struggles with very high-dimensional datasets due to the explosion of itemsets.
- **Maintenance:** Maintaining the vertical format for large datasets can be memory-intensive.

Complex and High Dimensionality